

RYAN PAUL TIRKEY

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EDUCATION

B.Tech - Computer Science & Engineering (Data Science) **Status:** Graduation this year
Dayananda Sagar College of Engineering | 2022-2026

PROJECTS

Local AI Research & Financial Analysis Platform | Python, FastAPI, Ollama, Qdrant, Tesseract OCR

- Designed prompts to extract and calculate 30+ financial metrics from unstructured annual reports into structured, schema-consistent CSV output — handling messy real-world formatting like parenthesis-negative numbers and missing values
- Built a semantic Q&A system over PDFs using local LLM + vector search, with prompts constrained to answer only from retrieved document context to prevent hallucination
- Built an OCR pipeline (Tesseract + OpenCV preprocessing) for scanned financial documents, automatically switching between direct text extraction and OCR depending on document type
- Ran all processing fully on-device — no data leaves the user's machine, relevant for handling sensitive documents

Automated Inbox Assistant | Java, LangChain4j, Ollama, Gmail API

- Engineered a zero-shot classification prompt distinguishing 6 email categories with explicit disambiguation rules for tricky edge cases (e.g., a recruiter's real outreach vs. a promotional email that just mentions "hiring")
- Built a two-stage pipeline — fast keyword rules for clear-cut cases, LLM only for ambiguous ones — to reduce compute cost while preserving accuracy on edge cases
- Designed a company-legitimacy prompt that combines web search results into a structured verdict + reasoning output, used to flag suspicious recruiter emails

Book Review Platform | Spring Boot, MongoDB, Gemini API

- Integrated Gemini API prompts to generate structured, on-demand book/author descriptions from stored review data
- Explored prompt-based review categorization during testing, evaluating LLM output consistency across different review text formats

TECHNICAL SKILLS

Languages: Java 21, Python

Backend & Frameworks: Spring Boot 3, FastAPI, REST APIs, MVC, LangChain4j, Streamlit

AI / ML: Ollama (Llama 3.2), LangChain4j, Qdrant, SentenceTransformers, Gemini API, Tesseract OCR, PyMuPDF, OpenCV

Databases: PostgreSQL, MongoDB, SQL

DevOps & Tools: Docker, docker-compose, Multi-stage Builds, Git/GitHub, Maven, Postman, Vercel

Security: Google OAuth 2.0 (Desktop Flow, Token Persistence, Credential Watcher)

Concepts: RAG, Microservices, OOP, REST, MVC, Vector Search, OCR Pipelines, Prompt Engineering, AI Tool Agents, Containerisation

INTERESTS

Learning new technologies · AI systems & LLM tooling · Game Development · Backend architecture